Big Data Analytics

Course #: 20262

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Course Description

This course is an introduction to large-scale data analytics. The Big Data course offers a concise exploration of the principles, technologies, and challenges associated with managing and analyzing large-scale datasets. Students will learn about the fundamentals of big data, including data acquisition, storage, processing, and analysis. Topics covered include distributed file systems, data modelling, data integration, parallel processing, and data visualization. Through practical exercises and projects, students will gain hands-on experience in working with big data tools and technologies, such as Apache Hadoop and Apache Spark. This course is essential for students interested in fields such as data science, data engineering, and business intelligence. Prerequisites: Basic understanding of programming and database concepts.

Course Prerequisites

- We expect you to have a solid background in Python programming and understand basic statistics and machine learning.
- This class includes topics from Data Analysis, and Machine Learning.
- To implement the assignments, students need to have excellent knowledge of the Python programming language and some basic Linux knowledge. Assignments are very time-consuming, and you should take this course when you have at least 7-15 hours per week.

Learning Objectives

By completing this course, you will be able to:

- Explain the main challenges of Big Data Processing.
- Run a Big Data Processing pipeline locally, and if the time and budget permit on Google Cloud (or Amazon AWS).
- Implement Big Data code in Apache Spark (in PySpark).
- Run Supervised and Unsupervised machine learning on Large-Scale Data.

Laptop Requirement

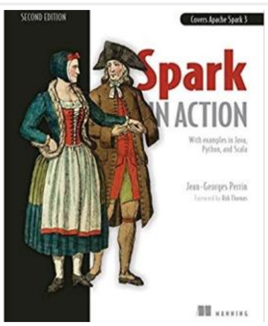
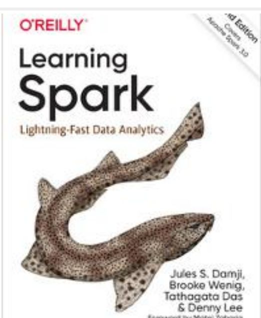
Students should have a personal laptop. We will use laptops to write Python programs and do the quizzes in the classroom. Also, for the Final exam, Laptops are required.

Materials

Required Book

There is no textbook required for the class. There are detailed lecture notes, and all class material will be conveyed during the lecture.

Recommended Books/ Other Materials and Resources

	Perrin, J. (2020). Spark in action (2nd ed.). (Covers Apache Spark 3 with examples in Java, Python, and Scala) O'Reilly Media Inc.
	Damji, J., Wenig, B., Das, T., Lee, D. (2020). Learning spark (2nd ed.) O'Reilly Media Inc.

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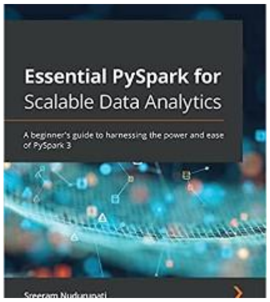
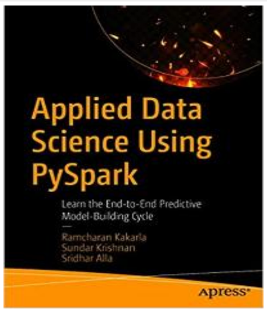
<tr><td>
</td><td>Nudurupati, S. (2021). Essential PySpark for scalable data analytics: A beginner's guide to harnessing the power and ease of PySpark 3
Packt Publishing</td></tr>
<tr><td>
</td><td>Ramcharan, K., Sundar, K., Alla, S. (2020). Applied data science using PySpark: Learn the end-to-end predictive model-building cycle
Apress</td></tr>
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APACHE Spark	Main Apache Spark documentation website

Course website

This course is accessible via UCAS google classroom.

Usage of Cloud Machines

In this class, we aim to use real-world cloud systems existing on Google Cloud (or Amazon AWS). You should never use your private account or use your credit card for this class assignment.

Class Meetings, Lectures & Assignments

Date	Module	Topics
	Module 1 Introduction to Big Data Processing	• Introduction to Big Data Analytics. What is Big Data? What are the challenges? • Introduction to Apache Hadoop and MapReduce. Apache Spark. • Spark programming. (Python and PySpark) • Spark - Resilient Distributed Dataset (RDDs).
	Module 2 Large-Scale Data Processing With PySpark	• Spark - RDDs, DataFrames, Spark SQL • PySpark + NumPy + SciPy, Code Optimization, Cluster Configurations • Linear Algebra Computation in Large Scale. • Distributed File Storage Systems
	Module 3 Data Modeling and Optimization Problems	• Introduction to modeling: numerical vs. probabilistic vs. Bayesian • Introduction to Optimization Problems • Batch and stochastic Gradient Descent • Newton's Method • Expectation-Maximization, • Markov Chain Monte Carlo (MCMC)
	Module 4 Large-Scale Supervised Learning	• Introduction to Supervised learning • Generalized Linear Models and Logistic Regression • Regularization • Support Vector Machine (SVM) and the kernel trick • Outlier Detection • Spark ML library
	Module 5 Large-Scale Unsupervised Learning	• Introduction to Unsupervised learning • K-means / K-medoids • Gaussian Mixture Models • Dimensionality Reduction • Spark MLlib for Unsupervised Learning
	Module 6 Large Scale Text Mining	• Latent Semantic Indexing • Topic models • Latent Dirichlet Allocation • Spark ML library for NLP

**The exact Modules topics, Lectures, Readings, and Assignments are subject to change and will be announced in class as applicable within a reasonable time frame.*

Good Luck